

Rafeek Homes Website

Step-by-step production installation on **Ubuntu 24.04 LTS** with **Nginx**

Stack: React (CRA) frontend · FastAPI (Python) backend · MongoDB · Nginx reverse proxy · systemd · HTTPS via Let's Encrypt

Overview & Architecture

This app has two parts served behind a single domain by Nginx:

- **Frontend** — a React app that is compiled to static files (`build/`). Nginx serves these directly.
- **Backend** — a FastAPI app run by `uvicorn` on `127.0.0.1:8001` . Every backend route starts with `/api` . Nginx forwards `/api` requests to it.
- **Database** — MongoDB running locally on `127.0.0.1:27017` .

```
Internet —> Nginx (:80 / :443)
              |
              |—— / —> React static files (/var/www/rafeek/build)
              |—— /api/... —> FastAPI (uvicorn @ 127.0.0.1:8001) —> MongoDB (:27017)
```

Before you start you need: a server running Ubuntu 24.04, a domain name (e.g. `rafeekhomes.com`) with its DNS A record pointed to your server's public IP, and SSH access with a `sudo` user.

1 Prepare the server

Connect over SSH and update the system:

```
ssh youruser@YOUR_SERVER_IP

sudo apt update && sudo apt upgrade -y
sudo apt install -y curl git build-essential ufw
```

Enable the firewall and allow SSH + web traffic:

```
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'
sudo ufw enable
sudo ufw status
```

2 Install Node.js 20 & Yarn (frontend tools)

```
# Node.js 20 LTS
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs

# Yarn (this project uses Yarn, NOT npm)
sudo corepack enable
corepack prepare yarn@1.22.22 --activate

node -v      # v20.x
yarn -v      # 1.22.x
```

3 Install Python 3 (backend runtime)

```
sudo apt install -y python3 python3-venv python3-pip
python3 --version # 3.12.x on Ubuntu 24.04
```

4 Install MongoDB Community 7.0

```
# Import the MongoDB public key
curl -fsSL https://pgp.mongodb.com/server-7.0.asc | \
  sudo gpg -o /usr/share/keyrings/mongodb-server-7.0.gpg --dearmor

# Add the repo (Ubuntu 24.04 "noble")
echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-7.0.gpg ]
https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/7.0 multiverse" | \
  sudo tee /etc/apt/sources.list.d/mongodb-org-7.0.list

sudo apt update
sudo apt install -y mongodb-org

# Start and enable on boot
sudo systemctl start mongod
sudo systemctl enable mongod
sudo systemctl status mongod --no-pager
```

MongoDB now listens on **127.0.0.1:27017** (localhost only) — perfect, the database should never be exposed to the internet.

5 Get the application code onto the server

Create an app directory and copy your project into it. Use **git** if your code is in a repository, or **scp** to upload it.

```

sudo mkdir -p /var/www/rafeek
sudo chown -R $USER:$USER /var/www/rafeek

# Option A - clone from git
git clone YOUR_REPO_URL /var/www/rafeek

# Option B - upload from your local machine (run this on your laptop)
# scp -r ./app/* youruser@YOUR_SERVER_IP:/var/www/rafeek/

```

After this you should have `/var/www/rafeek/backend` and `/var/www/rafeek/frontend` .

6 Configure & run the backend (FastAPI)

6.1 — Create a virtual environment and install dependencies

```

cd /var/www/rafeek/backend
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt

```

6.2 — Create the backend environment file

Create `/var/www/rafeek/backend/.env` with the production values:

```

MONGO_URL="mongodb://localhost:27017"
DB_NAME="rafeek_homes"
CORS_ORIGINS="https://rafeekhomes.com,https://www.rafeekhomes.com"

# --- Email: send the contact-form leads from YOUR mailbox (recommended) ---
# Google Workspace / Gmail example. SMTP_PASSWORD must be a 16-char Google App Password.
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_EMAIL=sales@sparkcurv.com
SMTP_PASSWORD=your_16_char_google_app_password
OWNER_EMAIL=sales@sparkcurv.com
EMAIL_FROM_NAME=Rafeek Homes

# --- Optional fallback only (leave blank/omit when self-hosting) ---
# EMERGENT_EMAIL_KEY=

```

Email is optional to boot. The backend starts fine even if no email is configured — inquiries are always saved to MongoDB. If `SMTP_*` is set, leads are emailed via your mailbox; `EMERGENT_EMAIL_KEY` is only a fallback used inside the Emergent platform and can be left out on your own server.

6.3 — Test it once by hand

```
cd /var/www/rafeek/backend
source venv/bin/activate
uvicorn server:app --host 127.0.0.1 --port 8001
# In another terminal: curl http://127.0.0.1:8001/api/
# Expected: {"message":"Rafeek Homes API"}
# Press Ctrl+C to stop.
```

6.4 — Run it permanently with systemd

Create the service file `/etc/systemd/system/rafeek-backend.service`:

```
[Unit]
Description=Rafeek Homes FastAPI backend
After=network.target mongod.service

[Service]
User=www-data
Group=www-data
WorkingDirectory=/var/www/rafeek/backend
EnvironmentFile=/var/www/rafeek/backend/.env
ExecStart=/var/www/rafeek/backend/venv/bin/uvicorn server:app --host 127.0.0.1 --port 8001 --workers 2
Restart=always
RestartSec=3

[Install]
WantedBy=multi-user.target
```

Fix permissions, then start the service:

```
sudo chown -R www-data:www-data /var/www/rafeek/backend
sudo systemctl daemon-reload
sudo systemctl enable rafeek-backend
sudo systemctl start rafeek-backend
sudo systemctl status rafeek-backend --no-pager

# View live logs anytime:
sudo journalctl -u rafeek-backend -f
```

7 Build the frontend (React)

7.1 — Point the frontend at your domain

Edit `/var/www/rafeek/frontend/.env` so the app calls your live domain:

```
REACT_APP_BACKEND_URL=https://rafeekhomes.com
```

Do **not** add a trailing slash and do **not** add `/api` here — the code appends `/api` itself. Nginx will route `/api` to the backend.

7.2 — Install dependencies and compile

```
cd /var/www/rafeek/frontend
yarn install
yarn build      # outputs static files into /var/www/rafeek/frontend/build
```

The `build/` folder now holds the production website Nginx will serve.

8 Configure Nginx

```
sudo apt install -y nginx
```

Create the site config `/etc/nginx/sites-available/rafeek`:

```
server {
    listen 80;
    server_name rafeekhomes.com www.rafeekhomes.com;

    root /var/www/rafeek/frontend/build;
    index index.html;

    # Gzip for faster loads
    gzip on;
    gzip_types text/plain text/css application/javascript application/json image/svg+xml;

    # API requests -> FastAPI backend
    location /api/ {
        proxy_pass http://127.0.0.1:8001;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 60s;
    }

    # Cache static assets aggressively
    location /static/ {
        expires 1y;
        add_header Cache-Control "public, immutable";
    }

    # React single-page-app routing: always fall back to index.html
    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

Enable the site and reload:

```
sudo ln -s /etc/nginx/sites-available/rafeek /etc/nginx/sites-enabled/  
sudo rm -f /etc/nginx/sites-enabled/default  
sudo nginx -t          # test config - must say "syntax is ok"  
sudo systemctl restart nginx
```

Visit <http://rafeekhomes.com> — the site should load.

9 Add free HTTPS (Let's Encrypt)

```
sudo apt install -y certbot python3-certbot-nginx  
  
sudo certbot --nginx -d rafeekhomes.com -d www.rafeekhomes.com  
# Follow the prompts (enter email, agree, choose redirect HTTP -> HTTPS).
```

Certbot edits your Nginx config for SSL automatically and sets up auto-renewal. Test renewal:

```
sudo certbot renew --dry-run
```

After HTTPS is live, make sure `REACT_APP_BACKEND_URL` uses `https://` (Step 7.1). If you changed it, rebuild the frontend with `yarn build`.

10 Verify the deployment

```
# Backend health  
curl https://rafeekhomes.com/api/  
  
# Contact form endpoint (saves + emails a lead)  
curl -X POST https://rafeekhomes.com/api/contact \  
  -H "Content-Type: application/json" \  
  -d '{"name":"Test","email":"test@example.com","phone":"9994556889","interest":"Buy a Home","message":"Hello"}'  
  
# List saved inquiries  
curl https://rafeekhomes.com/api/inquiries
```

Open the site in a browser, scroll through all sections, play a YouTube video, and submit the contact form — you should get a success toast.

Updating the site later

What changed	Commands to run
Frontend code	<code>cd /var/www/rafeek/frontend && git pull && yarn install && yarn build</code>
Backend code / requirements	<code>cd /var/www/rafeek/backend && git pull && source venv/bin/activate && pip install -r requirements.txt && sudo systemctl restart rafeek-backend</code>
Any <code>.env</code> value	Edit the file, then <code>sudo systemctl restart rafeek-backend</code> (backend) or <code>yarn build</code> (frontend)

Troubleshooting

Symptom	Check / Fix
502 Bad Gateway on <code>/api</code>	Backend is down. <code>sudo systemctl status rafeek-backend</code> and <code>sudo journalctl -u rafeek-backend -n 100</code> .
Site loads but API calls fail (CORS)	Make sure your domain is listed in <code>CORS_ORIGINS</code> in the backend <code>.env</code> , then restart the backend.
Blank page / 404 on refresh of a route	The <code>try_files ... /index.html</code> line in Nginx must be present (Step 8).
Contact emails not arriving	Inquiries still save to Mongo. Verify <code>EMERGENT_EMAIL_KEY</code> / email provider is reachable, or switch to your own SMTP/Resend.
Mongo connection error	<code>sudo systemctl status mongod</code> ; ensure <code>MONGO_URL=mongodb://localhost:27017</code> .
Nginx won't start	<code>sudo nginx -t</code> to see the config error line.

Security checklist (recommended)

- Keep the firewall on (Step 1) — only SSH + Nginx should be open.
- Never expose MongoDB to the internet; keep it bound to `127.0.0.1`.
- Store real secrets only in the server `.env` files (never commit them to git).
- Run `sudo apt update && sudo apt upgrade` regularly.
- Consider adding automated MongoDB backups (`mongodump` via cron).